CS19541 – COMPUTER NETWORKS LAB

PRACTICAL – 11

STATIC ROUTING AND RIP CONFIGURATION

Aim

To set up and test static routing and RIP routing between networks in Cisco Packet Tracer.

Requirements

Cisco Packet Tracer, routers, PCs, and suitable Ethernet and serial cables.

Procedure

1. Add three routers and the required PCs to the Packet Tracer workspace.
2. Connect all devices using the appropriate cables.
3. Configure IP addresses on router interfaces and PCs, including the correct default gateways.
4. Enable router interfaces with **no shutdown**. If a serial interface is the DCE side, set a clock rate when needed.
5. Add static routes with **ip route**; configure primary and backup paths where required.
6. Check static routing entries with **show ip route static**.
7. Start RIP using **router rip** and advertise directly connected networks with the **network** command.
8. Test communication with **ping** and **tracert**.

Sample Static Route Command

```
ip route [destination-network] [subnet-mask] [next-hop-IP]
```

Sample RIP Configuration

```
router rip
version 2
network [network-address]
no auto-summary
```

Input

Network addresses, subnet masks, next-hop IP addresses, and router interface configuration details.

Output

The routing tables show the configured static routes or the routes learned through RIP. The ping and tracert commands help confirm the route and connectivity between PCs.

Observation

1. Three routers and the required PCs were connected in Cisco Packet Tracer, and IP addresses were assigned to the interfaces and PCs.
2. Router interfaces were enabled with **no shutdown**. Suitable clock-rate and bandwidth values were configured on DCE serial interfaces as required.

3. Static routes, including primary and backup paths where needed, were added using **ip route** and checked with **show ip route static**.
4. RIP was activated using **router rip**, and directly connected networks were advertised using **network** commands.
5. Connectivity was checked with **ping** and **tracert**; communication between PC0 and PC1 was successfully established.

Result

Thus, static routing and RIP routing were configured and verified successfully in Cisco Packet Tracer.